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Kasper IoT / Telematics — Feature List

Prepared for: SUDO — scoping and architecture **Devices:** Teltonika FMC130 (heavy equipment, CAN/J1939) · FMC920 (light vehicles, OBD-II) **Phasing:** MVP = pilot deployment · P2 = post-pilot · P3 = analytics and AI layer

Data source column matters for scoping: **Device** features come free from the telemetry stream, **CAN** features depend on ECU access per machine, **Platform** features are application logic, and **AI/ML** features need accumulated history before they are useful.

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1. Live tracking and visibility

#	Feature	Source	Phase
1.1	Real-time asset position on map, all asset types	Device	MVP
1.2	Current status per asset — moving, idle, stopped, offline	Device (ignition + movement)	MVP
1.3	Last-known position with timestamp when offline	Platform	MVP
1.4	Historical route playback for a selected date range	Device	MVP
1.5	Trip history — start, end, distance, duration, stops	Device (odometer, ignition)	MVP
1.6	Asset list with filter by site, type, status, customer	Platform	MVP
1.7	Live map clustering for large fleets	Platform	P2
1.8	Street-level address resolution for positions	Platform (geocoding)	P2

2. Geofencing and site control

#	Feature	Source	Phase
2.1	Define site boundaries per project	Platform	MVP
2.2	Entry and exit alerts against a defined site	Device / Platform	MVP
2.3	Auto-geofence around a parked asset — theft detection	Device	MVP
2.4	Unauthorised movement alert — movement without ignition	Device (tow detect)	MVP
2.5	Time-on-site accumulation per asset per project	Platform	MVP
2.6	After-hours movement alert against a site schedule	Platform	P2
2.7	Route corridor deviation alerts for transport jobs	Platform	P2

3. Utilisation and billing evidence

The commercially critical module. Engine hours from the ECU are what settle billing disputes on long-term hire contracts.

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Note for architecture: 3.5 and 3.6 together are the reason tenancy must be pooled with dated assignment rather than siloed. A client must retain visibility of the period an asset was theirs and nothing outside it.

4. Fuel monitoring

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Scoping note: GNSS-estimated fuel (AVL IDs 12 and 13) is available on all devices without CAN, but it is an estimate. ECU-sourced fuel requires CAN access and is machine-dependent. Both should be represented distinctly in the data model — never merged into a single field.

5. Machine health and maintenance

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6. Driver and operator

#	Feature	Source	Phase
6.1	Harsh acceleration, braking and cornering detection	Device (accelerometer)	MVP
6.2	Over-speed alerts against configurable threshold	Device	MVP
6.3	Operator identification via iButton or RFID	Device (1-Wire)	P2

6.4	Driver scorecard and ranking	Platform	P2
6.5	Immobiliser — remote engine disable	Device (digital output)	P2
6.6	Crash detection and alert	Device	P2

Note on 6.5: technically available via the device's digital output, but carries obvious safety and liability considerations. Recommend legal review before enabling, and never permitting remote disable while an asset is in motion.

7. Alerts and notifications

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WhatsApp delivery is not optional for us — it is how our customers actually operate. It should be treated as a first-class channel in the architecture, not a later integration.

8. Multi-tenancy and access control

#	Feature	Source	Phase
8.1	Pooled multi-tenancy — shared platform, row-level isolation	Platform	MVP
8.2	Every record tagged by customer, project, site and asset	Platform	MVP
8.3	Date-ranged asset-to-customer assignment	Platform	MVP
8.4	Master user — visibility across all their projects	Platform	MVP
8.5	Site user — visibility limited to assigned	Platform	MVP

	site		
8.6	Vendor view — a fleet owner sees only their own assets	Platform	MVP
8.7	Internal Kasper admin — cross-tenant visibility	Platform	MVP
8.8	Role and permission management per customer	Platform	P2
8.9	Customer-branded portal	Platform	P3
8.10	Isolated deployment as a premium enterprise tier	Platform	P3

9. Reporting

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10. Device and fleet administration

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10.2 is a security control, not an admin convenience. The Teltonika protocol lets the server reject an unknown IMEI at handshake before any data is transmitted. That hook should be used.

11. Platform integration

#	Feature	Source	Phase
11.1	Telemetry surfaced in existing Kasper mobile and web app	Platform	MVP
11.2	Link telemetry to the existing job pipeline — dispatch through ePOD	Platform	MVP
11.3	GPS-verified arrival and departure on a job	Platform	MVP
11.4	ePOD tied to verified position and timestamp	Platform	P2
11.5	API for customer system integration	Platform	P2
11.6	Webhooks on telemetry events	Platform	P2

12. Analytics and AI

All P3. Dependent on accumulated telemetry history — not scopeable until the MVP has been running long enough to produce a usable dataset.

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MVP boundary

In scope for pilot: modules 1, 2, 3, 4 (device-sourced), 7 (7.1–7.3), 8 (8.1–8.7), 9 (9.1–9.5, 9.7), 10 (10.1–10.6), 11 (11.1–11.3), plus 5.3, 5.5, 5.6, 6.1, 6.2.

Deferred: operator ID, immobiliser, DTC codes, API and webhooks, scheduled reports, all of module 12.

The MVP boundary is drawn so that everything in it is provable at 50–100 devices and directly supports the commercial case — utilisation billing, theft prevention and site visibility. Module 12 is genuine long-term value but needs a data history that doesn't exist yet, and should not carry cost or scope in the pilot estimate.

Two things for SUDO to note

Module 3 drives the architecture. Engine hours as billing evidence is why we need tamper-evident storage, dated tenant assignment, and an ingestion path that never acknowledges a record before it is durably stored. It is not a reporting feature.

Module 8 is settled. Pooled multi-tenancy with row-level isolation. Our customers are many and small, and our assets are rented and change hands. Per-tenant infrastructure does not fit either fact.